Why Does Android Run Time (ART) Score Over Dalvik?

This article describes the advantages of using the Android Run Time (ART) over Dalvik VM in Android devices.

Android [4.4] KitKat users must have noticed the new option to choose the default runtime environment in Android (Dalvik or ART). Dalvik has been the runtime environment in Android since the very beginning. Although ART was added as an experimental feature in KitKat, it has now replaced Dalvik from Android [5.0] Lollipop onwards.

What is a runtime environment?

When a software program is executed, it is in a runtime state, during which the program sends instructions to the computer's processor to access the system's resources. To do this, we have a runtime environment that executes software instructions when a program is running. These instructions translate the software code into machine code (byte code) that the computer is capable of understanding. In simpler terms, it means that all Android application files (APKs) are basically uncompiled instructions.

Why does Android use a virtual machine?

Android uses a virtual machine as its runtime environment to compile and run its applications. Unlike our Virtual Box, this virtual machine does not emulate the entire computer! Using a virtual machine ensures that the application execution is isolated from the core operating system, so even if the application contains some malicious code, it cannot directly affect the system. This provides stability and reliability for the operating system. This also provides more compatibility—since different processors use different instruction sets and architectures, compilation on the device ensures compatibility with the specific device.

How does Dalvik work? Why is it being replaced?

Dalvik uses a JIT (Just-In-Time) compiler for its virtual machine. Applications need a lot of resources to run. Taking up a lot of resources can slow down the system. With